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ESTIMATES FOR A CERTAIN BILINEAR FOURIER INTEGRAL OPERATOR

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ABSTRACT. In this paper, we consider the boundedness from $H^1 \times L^\infty$ to L^1 of bilinear Fourier integral operators with non-degenerate phase functions and amplitudes in $BS^{-n/2}_{1,0}$. Our result gives an improvement of Rodríguez-López, Rule, and Staubach's result.

1. INTRODUCTION

The (linear) Fourier integral operator T^ϕ_σ treated in this paper is the following form:

$$T^\phi_\sigma f(x) = \int_{\mathbb{R}^n} e^{i\phi(x,\xi)}\sigma(x,\xi)\widehat{f}(\xi)\,d\xi, \quad x \in \mathbb{R}^n,$$

for f in the Schwartz class $\mathcal{S}(\mathbb{R}^n)$, where $\widehat{f}$ denotes the Fourier transform. The function ϕ is called the *phase function*, usually assumed homogeneity and non-degeneracy conditions, and the function σ is called the *amplitude*, usually taken to be in some Hörmander class (see Definitions 1.1 and 1.2 below). For the boundedness of the operator T^ϕ_σ, we use the following terminology. Let X and Y be function spaces on $\mathbb{R}^n$ equipped with quasi-norms or seminorms $\|\cdot\|_X$ and $\|\cdot\|_Y$, respectively. If there exists a constant C such that

$$\|T^\phi_\sigma f\|_Y \le C\|f\|_X \tag{1.1}$$

holds for all $f \in \mathcal{S} \cap X$, then we say that T^ϕ_σ is bounded from X to Y. The smallest constant C of (1.1) is denoted by $\|T^\phi_\sigma\|_{X\to Y}$.

In order to illustrate boundedness results of the operator T^ϕ_σ, we recall the following definitions for phase functions and amplitudes.

Definition 1.1. A function $\phi(x,\xi) \in C^\infty(\mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\}))$ is called a non-degenerate phase function, if it is real-valued, positively homogeneous of degree one in the variable ξ, *i.e.*,

$$\phi(x,t\xi) = t\phi(x,\xi), \quad t > 0, \ \ x \in \mathbb{R}^n, \ \ \xi \in \mathbb{R}^n \setminus \{0\},$$

and satisfies the *non-degeneracy condition*

$$\det\Big(\frac{\partial^2\phi(x,\xi)}{\partial x \partial \xi}\Big) \neq 0$$

for x in the spatial support of the amplitude and for $\xi \in \mathbb{R}^n \setminus \{0\}$.

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Definition 1.2. Let $m \in \mathbb{R}$ and $N \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}$. The class $S^m_{1,0}(N)$ is defined to be the set of all C^∞ functions $\sigma = \sigma(x,\xi)$ on $\mathbb{R}^{2n}$ that satisfy the estimate

$$\left|\partial_x^\alpha \partial_\xi^\beta \sigma(x,\xi)\right| \le C_{\alpha,\beta}\,(1+|\xi|)^{m-|\beta|}$$

for all multi-indices satisfying $|\alpha|, |\beta| \le N$. We define

$$\|\sigma\|_{S^m_{1,0}(N)} = \max_{|\alpha|,|\beta|\le N} \left(\sup_{x,\xi\in\mathbb{R}^n} (1+|\xi|)^{-(m-|\beta|)} \left|\partial_x^\alpha \partial_\xi^\beta \sigma(x,\xi)\right| \right).$$

Now, we recall the following theorem.

Theorem A. *Let ϕ be a non-degenerate phase function and an amplitude $\sigma(x,\xi)$ be compactly supported in the spatial variable x. Then, if $1 < p < \infty$ and $m = -(n-1)|1/p - 1/2|$, or if $1 \le p \le \infty$ and $m < -(n-1)|1/p - 1/2|$, there exist a positive constant C and a positive integer N such that*

$$\|T^\phi_\sigma\|_{L^p \to L^p} \le C \|\sigma\|_{S^m_{1,0}(N)}.$$

The first assertion was proved by Hörmander [Hör] for $p = 2$ and Seeger–Sogge–Stein [SSS] for $1 < p < \infty$, and the second by Dos Santos Ferreira–Staubach [DS, Theorems 2.1, 2.2, and 2.16]. It is worth mentioning that the number $-(n-1)|1/p - 1/2|$ is optimal (see [M]).

Based on the above result, we shall consider a bilinear analogue. The bilinear Fourier integral operator T^{ϕ_1,ϕ_2}_σ considered in this paper is defined by, for non-degenerate phase functions ϕ_1 and ϕ_2 and for an amplitude $\sigma \in L^\infty(\mathbb{R}^{3n})$,

$$T^{\phi_1,\phi_2}_\sigma(f,g)(x) = \iint_{\mathbb{R}^n\times\mathbb{R}^n} e^{i\phi_1(x,\xi)+i\phi_2(x,\eta)} \sigma(x,\xi,\eta) \widehat{f}(\xi)\widehat{g}(\eta)\, d\xi d\eta, \quad x \in \mathbb{R}^n,$$

for $f, g \in \mathcal{S}(\mathbb{R}^n)$. If X, Y, and Z are function spaces on $\mathbb{R}^n$ equipped with quasi-norms or seminorms $\|\cdot\|_X$, $\|\cdot\|_Y$, and $\|\cdot\|_Z$, respectively, and if there exists a constant C such that

$$\|T^{\phi_1,\phi_2}_\sigma(f,g)\|_Z \le C \|f\|_X \|g\|_Y \tag{1.2}$$

holds for all $f \in \mathcal{S} \cap X$ and $g \in \mathcal{S} \cap Y$, then we say that T^{ϕ_1,ϕ_2}_σ is bounded from $X \times Y$ to Z. The smallest constant C of (1.2) is denoted by $\|T^{\phi_1,\phi_2}_\sigma\|_{X\times Y\to Z}$. For the functions spaces X, Y, and Z, we consider the Lebesgue space L^p, the Hardy space H^p, the local Hardy space h^p, and the space BMO. We note that $H^p = h^p = L^p$ if $1 < p \le \infty$ and $H^1 \hookrightarrow h^1 \hookrightarrow L^1$. For H^p and BMO, see, *e.g.,* [S, Chapters III and IV] and, for h^p, see [Gol]. In the special case when $\phi_i(x,\xi) = x \cdot \xi$, $i = 1, 2$, the bilinear Fourier integral operator is called a bilinear pseudo-differential operator, which is the operator originated by Coifman–Meyer [CM], and now vastly wide study has been done (see, *e.g.*, [BBMNT, BT, GT]).

Now, in order to state the boundedness result for the bilinear operator T^{ϕ_1,ϕ_2}_σ, we define the bilinear Hörmander class as follows.

Definition 1.3. For $m \in \mathbb{R}$, the class $BS^m_{1,0}$ is defined to be the set of all C^∞ functions $\sigma = \sigma(x,\xi,\eta)$ on $\mathbb{R}^{3n}$ that satisfy the estimate

$$\left|\partial_x^\alpha \partial_\xi^\beta \partial_\eta^\gamma \sigma(x,\xi,\eta)\right| \le C_{\alpha,\beta,\gamma}\,(1+|\xi|+|\eta|)^{m-|\beta|-|\gamma|}$$

for all multi-indices $\alpha, \beta, \gamma \in (\mathbb{N}_0)^n$.

For this class, the following theorem holds.

Theorem B. *Let $m \in \mathbb{R}$, $n \geq 2$, $1 \leq p, q \leq \infty$, and $1/p + 1/q = 1/r$. Suppose that ϕ_1 and ϕ_2 are non-degenerate phase functions and an amplitude $\sigma \in BS^m_{1,0}$ is compactly supported in the spatial variable. Then, if*

$$m = m_0(p,q) := -(n-1)\Big(\Big|\frac{1}{p} - \frac{1}{2}\Big| + \Big|\frac{1}{q} - \frac{1}{2}\Big|\Big),$$

there exists a positive constant C such that

$$\|T^{\phi_1,\phi_2}_\sigma\|_{H^p \times H^q \to L^r} \leq C,$$

where L^r should be replaced by BMO when $r = \infty$.

The boundedness property for the bilinear operator T^{ϕ_1,ϕ_2}_σ was first investigated by Grafakos–Peloso [GP], where the above-mentioned boundedness for $1 < p, q < 2$ and $m < m_0(p,q)$ was obtained. After that, Rodríguez-López–Rule–Staubach [RRS1] generalized it to the full range of $1 \leq p, q \leq \infty$, and gave Theorem B. Also, we remark that, in [RS], Rodríguez-López–Staubach considered the bilinear operator T^{ϕ_1,ϕ_2}_σ for an amplitude without compactness of the spatial support and for phase functions with a bit stronger condition, and gave a global boundedness from $L^p \times L^q$ to L^r when $m < m_0(p,q)$. In [RRS2], Rodríguez-López–Rule–Staubach obtained a global boundedness from $L^2 \times L^2$ to L^1 for $m = m_0(2,2) = 0$.

Quite recently, for a typical case of phase functions derived from the wave operators and for an x-independent amplitude $\sigma = \sigma(\xi, \eta)$, the authors [KMT] improved the order m stated in Theorem B. To describe the result obtained there, we define

$$m_1(p,q) = \begin{cases} -(n-1)\big(|\frac{1}{p} - \frac{1}{2}| + |\frac{1}{q} - \frac{1}{2}|\big) & \text{if } 1 \leq p, q \leq 2 \text{ or if } 2 \leq p, q \leq \infty, \\ -\big(\frac{1}{p} - \frac{1}{2}\big) - (n-1)\big(\frac{1}{2} - \frac{1}{q}\big) & \text{if } 1 \leq p \leq 2 \leq q \leq \infty \text{ and } \frac{1}{p} + \frac{1}{q} \leq 1, \\ -(n-1)\big(\frac{1}{p} - \frac{1}{2}\big) - \big(\frac{1}{2} - \frac{1}{q}\big) & \text{if } 1 \leq p \leq 2 \leq q \leq \infty \text{ and } \frac{1}{p} + \frac{1}{q} \geq 1, \\ -(n-1)\big(\frac{1}{2} - \frac{1}{p}\big) - \big(\frac{1}{q} - \frac{1}{2}\big) & \text{if } 1 \leq q \leq 2 \leq p \leq \infty \text{ and } \frac{1}{p} + \frac{1}{q} \leq 1, \\ -\big(\frac{1}{2} - \frac{1}{p}\big) - (n-1)\big(\frac{1}{q} - \frac{1}{2}\big) & \text{if } 1 \leq q \leq 2 \leq p \leq \infty \text{ and } \frac{1}{p} + \frac{1}{q} \geq 1. \end{cases}$$

With this notation, the following global boundedness holds.

Theorem C. *Let $m \in \mathbb{R}$, $n \geq 2$, $1 \leq p, q \leq \infty$, and $1/p + 1/q = 1/r$. Suppose that $\phi_i(x,\xi) = x \cdot \xi + |\xi|$, $i = 1, 2$, and $\sigma = \sigma(\xi, \eta) \in BS^m_{1,0}$. Then, if $m = m_1(p,q)$, there exists a positive constant C such that*

$$\|T^{\phi_1,\phi_2}_\sigma\|_{H^p \times H^q \to L^r} \leq C,$$

where L^r should be replaced by BMO when $r = \infty$.

Since $m_0(p,q) \leq m_1(p,q)$ for $n \geq 2$ and $1 \leq p, q \leq \infty$, we see that Theorem C is an improvement of Theorem B. Also, it should be remarked that the number $m_1(p,q)$ is optimal when $1 \leq p, q \leq 2$, $2 \leq p, q \leq \infty$, or $1/p + 1/q = 1$ (see [KMT, Theorem 1.5]).

The purpose of this paper is to improve Theorem B and to generalize Theorem C to the boundedness for bilinear Fourier integral operators. The main theorem reads as follows.

Theorem 1.4. *Let $m \in \mathbb{R}$, $n \geq 2$, $1 \leq p, q \leq \infty$, and $1/p + 1/q = 1/r$. Suppose that ϕ_1 and ϕ_2 are non-degenerate phase functions and an amplitude $\sigma \in BS^m_{1,0}$ is compactly supported in the spatial variable. Then, if $m = m_1(p,q)$, there exists a positive constant C such that*

$$\|T^{\phi_1,\phi_2}_\sigma\|_{H^p \times H^q \to L^r} \leq C,$$

where L^r should be replaced by BMO when $r = \infty$.

The contents of the rest of the paper are as follows. In Section 2, we present and prove Proposition 2.1 concerning the boundedness from $h^1 \times L^\infty$ to L^1, which is an endpoint case of Theorem 1.4 and implies Theorem 1.4 with the aid of complex interpolation. In Appendix A, we show Lemma 2.2 to be used for proving Proposition 2.1.

We end this section by introducing some notations used throughout this paper.

Notation 1.5. The Fourier transform and the inverse Fourier transform on $\mathbb{R}^n$ are defined by

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} e^{-i\xi\cdot x} f(x)\, dx,$$
$$(g)^\vee(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\xi\cdot x} g(\xi)\, d\xi.$$

We take $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$ satisfying that $\varphi = 1$ on $\{|\xi| \le 1\}$, $\operatorname{supp}\varphi \subset \{|\xi| \le 2\}$, $\operatorname{supp}\psi \subset \{1/2 \le |\xi| \le 2\}$, and $\varphi + \sum_{j\in\mathbb{N}} \psi(2^{-j}\cdot) = 1$. In what follows, we will write $\psi_0 = \varphi$, $\psi_j = \psi(2^{-j}\cdot)$ for $j \in \mathbb{N}$, and $\varphi_j = \varphi(2^{-j}\cdot)$ for $j \in \mathbb{N}_0$. Then, we see that $\varphi_0 = \psi_0 = \varphi$ and

$$\sum_{j=0}^{k} \psi_j = \varphi_k, \quad k \in \mathbb{N}_0.$$

We define C^∞ function $\zeta = 1 - \varphi$. Then we have $\partial^\alpha \zeta \in C_0^\infty(\mathbb{R}^n)$ for $|\alpha| \ge 1$, $\zeta = \sum_{j\in\mathbb{N}} \psi_j$, and

$$\zeta = 0 \ \text{ on } \ \{|\xi| \le 1\}, \quad \zeta = 1 \ \text{ on } \ \{|\xi| \ge 2\}.$$

For a smooth function θ on $\mathbb{R}^n$ and for $N \in \mathbb{N}_0$, we write $\|\theta\|_{C^N} = \max_{|\alpha|\le N} \sup_\xi \left|\partial_\xi^\alpha \theta(\xi)\right|$.

2. Proof of Theorem 1.4

By complex interpolation, Theorem 1.4 follows from Theorem B and Proposition 2.1 below, and thus, this section is only devoted to proving the following statement.

Proposition 2.1. *Let $n \ge 2$. Suppose that ϕ_1 and ϕ_2 are non-degenerate phase functions and an amplitude $\sigma \in BS^{-n/2}_{1,0}$ is compactly supported in the spatial variable. Then, there exists a positive constant C such that*

$$\|T^{\phi_1,\phi_2}_\sigma\|_{h^1\times L^\infty \to L^1} \le C.$$

Proof. Before starting the proof, we give a remark on the non-degeneracy conditions of ϕ_1 and ϕ_2. Let K be the closure of the set $\{x \in \mathbb{R}^n \mid \sigma(x,\cdot,\cdot) \ne 0\}$. Then, since ϕ_1 and ϕ_2 satisfy the non-degeneracy conditions for $x \in K$ and $\xi \in \mathbb{R}^n \setminus \{0\}$, we see from continuity that there exists a neighborhood $\widetilde{K}$ of the set K such that

$$\det\Big(\frac{\partial^2\phi_i(x,\xi)}{\partial x \partial \xi}\Big) \ne 0 \ \text{ for } \ x \in \widetilde{K}, \ \ \xi \in \mathbb{R}^n \setminus \{0\}, \ \ i = 1, 2. \tag{2.1}$$

In the following argument, $\widetilde{K}$ always denotes such a neighborhood of K.

Now, we shall begin with the proof of the proposition. We first decompose σ as

$$\sigma(x,\xi,\eta) = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} \sigma(x,\xi,\eta)\psi_j(\xi)\psi_k(\eta)$$

$$
\begin{aligned}
&= \sigma(x,\xi,\eta)\varphi(\xi)\varphi(\eta) + \sum_{j=1}^{\infty}\sum_{k=0}^{j}\sigma(x,\xi,\eta)\psi_j(\xi)\psi_k(\eta) + \sum_{k=1}^{\infty}\sum_{j=0}^{k-1}\sigma(x,\xi,\eta)\psi_j(\xi)\psi_k(\eta)\\
&= \sigma(x,\xi,\eta)\varphi(\xi)\varphi(\eta) + \sum_{j\in\mathbb{N}}\sigma(x,\xi,\eta)\psi_j(\xi)\varphi_j(\eta) + \sum_{k\in\mathbb{N}}\sigma(x,\xi,\eta)\varphi_{k-1}(\xi)\psi_k(\eta)\\
&= \sigma_0(x,\xi,\eta) + \sigma_{\mathrm{I}}(x,\xi,\eta) + \sigma_{\mathrm{II}}(x,\xi,\eta).
\end{aligned}
$$

We consider the amplitude σ_{I}. Take a function $\chi \in C_0^\infty(\mathbb{R}^n)$ such that

$$\chi = 1 \ \text{ on } \ K, \quad \operatorname{supp}\chi \subset \widetilde{K},$$

and functions $\widetilde{\psi}, \widetilde{\varphi} \in C_0^\infty(\mathbb{R}^n)$ such that

$$
\begin{aligned}
&\widetilde{\psi} = 1 \ \text{ on } \ \{2^{-1} \le |\xi| \le 2\}, && \operatorname{supp}\widetilde{\psi} \subset \{3^{-1} \le |\xi| \le 3\},\\
&\widetilde{\varphi} = 1 \ \text{ on } \ \{|\xi| \le 2\}, && \operatorname{supp}\widetilde{\varphi} \subset \{|\xi| \le 3\}.
\end{aligned}
$$

Then we can write

$$\sigma_{\mathrm{I}}(x,\xi,\eta) = \sum_{j\in\mathbb{N}}\sigma(x,\xi,\eta)\widetilde{\psi}(2^{-j}\xi)\widetilde{\varphi}(2^{-j}\eta)\,\chi(x)\psi_j(\xi)\chi(x)\varphi_j(\eta).$$

Since $\sigma \in BS^{-n/2}_{1,0}$ and

$$\operatorname{supp}\sigma(x,2^j\xi,2^j\eta)\widetilde{\psi}(\xi)\widetilde{\varphi}(\eta) \subset K \times \{3^{-1} \le |\xi| \le 3\} \times \{|\eta| \le 3\},$$

the following estimate holds:

$$\left|\partial_\xi^\alpha\partial_\eta^\beta\big(\sigma(x,2^j\xi,2^j\eta)\widetilde{\psi}(\xi)\widetilde{\varphi}(\eta)\big)\right| \le C_{\alpha,\beta}\,2^{-jn/2}$$

with $C_{\alpha,\beta}$ independent of $j \in \mathbb{N}_0$. Hence, by the Fourier series expansion with respect to the variables ξ and η, we can write

$$\sigma(x,2^j\xi,2^j\eta)\widetilde{\psi}(\xi)\widetilde{\varphi}(\eta) = \sum_{a,b\in\mathbb{Z}^n} c_{\mathrm{I},j}^{(a,b)}(x)e^{ia\cdot\xi}e^{ib\cdot\eta}, \quad |\xi| < \pi, \ \ |\eta| < \pi,$$

with the coefficient satisfying that for $L > 0$

$$|c_{\mathrm{I},j}^{(a,b)}(x)| \lesssim 2^{-jn/2}(1+|a|)^{-L}(1+|b|)^{-L}, \quad x \in \mathbb{R}^n. \tag{2.2}$$

Changing variables $\xi \to 2^{-j}\xi$ and $\eta \to 2^{-j}\eta$ and multiplying $(\chi(x))^2\psi_j(\xi)\varphi_j(\eta)$, we obtain

$$\sigma(x,\xi,\eta)\psi_j(\xi)\varphi_j(\eta) = \sum_{a,b\in\mathbb{Z}^n} c_{\mathrm{I},j}^{(a,b)}(x)\,e^{ia\cdot 2^{-j}\xi}e^{ib\cdot 2^{-j}\eta}\chi(x)\psi_j(\xi)\chi(x)\varphi_j(\eta).$$

Hence, by the definitions of ψ_j and φ_j in Notation 1.5, the amplitude σ_{I} is written as

$$
\begin{aligned}
\sigma_{\mathrm{I}}(x,\xi,\eta) &= \sum_{a,b\in\mathbb{Z}^n}\sum_{j\in\mathbb{N}} c_{\mathrm{I},j}^{(a,b)}(x)\,e^{ia\cdot 2^{-j}\xi}e^{ib\cdot 2^{-j}\eta}\chi(x)\psi(2^{-j}\xi)\,\chi(x)\varphi(2^{-j}\eta)\\
&= \sum_{a,b\in\mathbb{Z}^n}\sigma_{\mathrm{I}}^{(a,b)}(x,\xi,\eta),
\end{aligned}
$$

where

$$\sigma_{\mathrm{I}}^{(a,b)}(x,\xi,\eta) = \sum_{j\in\mathbb{N}} c_{\mathrm{I},j}^{(a,b)}(x)\,\Psi^{(a)}(x,2^{-j}\xi)\,\Phi^{(b)}(x,2^{-j}\eta)$$

and

$$\Psi^{(\nu)}(x,\xi) = \chi(x)\,e^{i\nu\cdot\xi}\psi(\xi), \quad \Phi^{(\nu)}(x,\eta) = \chi(x)\,e^{i\nu\cdot\eta}\varphi(\eta), \quad \nu \in \mathbb{Z}^n.$$

By similar arguments, σ_0 can be written as

$$\sigma_0(x,\xi,\eta)=\sum_{a,b\in\mathbb{Z}^n} c_0^{(a,b)}(x)\,\Phi^{(a)}(x,\xi)\,\Phi^{(b)}(x,\eta),$$

where the coefficient $c_0^{(a,b)}$ satisfies the same condition as in (2.2) with $j=0$, and σ_{II} can be written as

$$\sigma_{\mathrm{II}}(x,\xi,\eta)=\sum_{a,b\in\mathbb{Z}^n}\sum_{j\in\mathbb{N}} c_{\mathrm{II},j}^{(a,b)}(x)\,\Phi^{(a)}(x,2^{-(j-1)}\xi)\,\Psi^{(b)}(x,2^{-j}\eta),$$

where the coefficient $c_{\mathrm{II},j}^{(a,b)}$ satisfies the same condition as in (2.2).

Hereafter we shall consider the amplitudes $\widetilde{\sigma}_0$ and $\widetilde{\sigma}$ of the forms

$$\widetilde{\sigma}_0(x,\xi,\eta)=c_0(x)\,\Theta_1(x,\xi)\,\Theta_2(x,\eta), \tag{2.3}$$

$$\widetilde{\sigma}(x,\xi,\eta)=\sum_{j\in\mathbb{N}} c_j(x)\,\Theta_1(x,2^{-j}\xi)\,\Theta_2(x,2^{-j}\eta), \tag{2.4}$$

where $(c_j(x))_{j\in\mathbb{N}_0}$ is a sequence of complex-valued functions satisfying

$$|c_j(x)|\le 2^{-jn/2}A,\quad x\in\mathbb{R}^n,\ \ j\in\mathbb{N}_0, \tag{2.5}$$

with some $A\in(0,\infty)$, and Θ_1 and Θ_2 are smooth functions on $\mathbb{R}^{2n}$ given by

$$\Theta_1(x,\xi)=\chi(x)\theta_1(\xi),\quad \Theta_2(x,\eta)=\chi(x)\theta_2(\eta) \tag{2.6}$$

with $\chi,\theta_1,\theta_2\in\mathcal{S}(\mathbb{R}^n)$ satisfying that

$$\operatorname{supp}\chi\subset\widetilde{K},\quad \operatorname{supp}\theta_1,\operatorname{supp}\theta_2\subset\{|\xi|\le 2\} \tag{2.7}$$

(for the set $\widetilde{K}$, see the argument around (2.1)). For such $\widetilde{\sigma}_0$ and $\widetilde{\sigma}$, we shall prove that there exist constants $c\in(0,\infty)$ and $N\in\mathbb{N}$ such that

$$\|T^{\phi_1,\phi_2}_{\widetilde{\sigma}_0}\|_{h^1\times L^\infty\to L^1},\ \|T^{\phi_1,\phi_2}_{\widetilde{\sigma}}\|_{h^1\times L^\infty\to L^1}\le cA\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}. \tag{2.8}$$

If this is proved, by applying (2.8) for $\widetilde{\sigma}$ to $c_j=c_{\mathrm{I},j}^{(a,b)}$, $\Theta_1=\Psi^{(a)}$, and $\Theta_2=\Phi^{(b)}$, we have

$$\begin{aligned}\|T^{\phi_1,\phi_2}_{\sigma_{\mathrm{I}}^{(a,b)}}\|_{h^1\times L^\infty\to L^1}&\lesssim(1+|a|)^{-L}(1+|b|)^{-L}\|e^{ia\cdot\xi}\psi(\xi)\|_{C^N_\xi}\|e^{ib\cdot\eta}\varphi(\eta)\|_{C^N_\eta}\\&\lesssim(1+|a|)^{-L+N}(1+|b|)^{-L+N},\end{aligned}$$

and, thus, taking L sufficiently large and taking sum over $a,b\in\mathbb{Z}^n$, we obtain

$$\big\|T^{\phi_1,\phi_2}_{\sigma_{\mathrm{I}}}\big\|_{h^1\times L^\infty\to L^1}\lesssim 1.$$

In the same way, we obtain

$$\|T^{\phi_1,\phi_2}_{\sigma_0}\|_{h^1\times L^\infty\to L^1},\ \|T^{\phi_1,\phi_2}_{\sigma_{\mathrm{II}}}\|_{h^1\times L^\infty\to L^1}\lesssim 1.$$

The above three estimates complete the proof of the proposition.

Thus the proof is reduced to showing (2.8) for $\widetilde{\sigma}_0$ and $\widetilde{\sigma}$ given by (2.3)–(2.7). Using the smooth functions φ and ζ of Notation 1.5, we decompose the amplitude $\widetilde{\sigma}$ into two parts:

$$\begin{aligned}&\widetilde{\sigma}(x,\xi,\eta)=\tau_1(x,\xi,\eta)+\tau_2(x,\xi,\eta),\\&\tau_1(x,\xi,\eta)=\varphi(\eta)\widetilde{\sigma}(x,\xi,\eta),\quad \tau_2(x,\xi,\eta)=\zeta(\eta)\widetilde{\sigma}(x,\xi,\eta).\end{aligned}$$

Then, bilinear Fourier integral operators with the amplitudes $\widetilde{\sigma}_0, \tau_1, \tau_2$ are expressed, by using linear ones, as

$$T^{\phi_1,\phi_2}_{\widetilde{\sigma}_0}(f,g)(x) = c_0(x)\, T^{\phi_1}_{\Theta_1} f(x)\; T^{\phi_2}_{\Theta_2} g(x), \tag{2.9}$$

$$T^{\phi_1,\phi_2}_{\tau_1}(f,g)(x) = \sum_{j\in\mathbb{N}} c_j(x)\, T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)} f(x)\; T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\varphi(\eta)} g(x), \tag{2.10}$$

$$T^{\phi_1,\phi_2}_{\tau_2}(f,g)(x) = \sum_{j\in\mathbb{N}} c_j(x)\, T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)} f(x)\; T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)} g(x). \tag{2.11}$$

We shall consider the amplitudes and the phase functions of the linear Fourier integral operators appearing above. For the amplitudes, we observe that, if $m \le 0$ and $N \in \mathbb{N}_0$, then

$$\begin{aligned}
&\|\Theta_i(x,2^{-j}\xi)\|_{S^m_{1,0}(N)} = \|\chi(x)\,\theta_i(2^{-j}\xi)\|_{S^m_{1,0}(N)} \lesssim 2^{-jm}\|\theta_i\|_{C^N}, \quad i=1,2,\\
&\|\Theta_2(x,2^{-j}\eta)\varphi(\eta)\|_{S^m_{1,0}(N)} = \|\chi(x)\,\theta_2(2^{-j}\eta)\varphi(\eta)\|_{S^m_{1,0}(N)} \lesssim \|\theta_2\|_{C^N},\\
&\|\Theta_2(x,2^{-j}\eta)\zeta(\eta)\|_{S^m_{1,0}(N)} = \|\chi(x)\,\theta_2(2^{-j}\eta)\zeta(\eta)\|_{S^m_{1,0}(N)} \lesssim 2^{-jm}\|\theta_2\|_{C^N}
\end{aligned}$$

hold for all $j \in \mathbb{N}_0$. In fact, since $|\xi| \le 2^{j+1}$ holds on the support of $\theta_i(2^{-j}\cdot)$, if $m \le 0$ and $\beta \in (\mathbb{N}_0)^n$, then we have

$$(1+|\xi|)^{-(m-|\beta|)} \lesssim 2^{-j(m-|\beta|)}, \quad j \in \mathbb{N}_0,$$

which implies the first inequality. By using the facts that φ and $\partial^\alpha\zeta$, $|\alpha| \ge 1$, belong to $C^\infty_0(\mathbb{R}^n)$, the remaining two estimates can be obtained. For the phase functions, we see from (2.1), (2.6), and (2.7) that $\phi_i(x,\xi)$, $i=1,2$, satisfies the non-degeneracy condition for x in the spatial support of $\Theta_i(x,2^{-j}\xi)$ and for $\xi \in \mathbb{R}^n\setminus\{0\}$. Moreover, from the assumption, they are real-valued and positively homogeneous of degree one in the frequency variable.

Hence, we can apply Theorem A to the linear Fourier integral operators in (2.9), (2.10), and (2.11): if

$$m_\epsilon = -\frac{n-1}{2} - \epsilon, \quad 0<\epsilon<1/2,$$

then there exists $N \in \mathbb{N}$ such that

$$\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}\|_{L^1\to L^1} \lesssim 2^{-jm_\epsilon}\|\theta_1\|_{C^N}, \tag{2.12}$$

$$\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}\|_{L^2\to L^2} \lesssim \|\theta_1\|_{C^N}, \tag{2.13}$$

$$\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)}\|_{L^\infty\to L^\infty} \lesssim 2^{-jm_\epsilon}\|\theta_2\|_{C^N}, \tag{2.14}$$

$$\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\varphi(\eta)}\|_{L^\infty\to L^\infty} \lesssim \|\theta_2\|_{C^N}, \tag{2.15}$$

$$\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)}\|_{L^2\to L^2} \lesssim \|\theta_2\|_{C^N}, \tag{2.16}$$

$$\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)}\|_{L^\infty\to L^\infty} \lesssim 2^{-jm_\epsilon}\|\theta_2\|_{C^N} \tag{2.17}$$

hold for all $j \in \mathbb{N}_0$.

Now, we shall actually estimate the bilinear operators in (2.9), (2.10), and (2.11). Estimates for $\widetilde{\sigma}_0$ and τ_1 are simple. In fact, we have by (2.9), (2.12), and (2.14) for $j=0$

$$\begin{aligned}
\|T^{\phi_1,\phi_2}_{\widetilde{\sigma}_0}(f,g)\|_{L^1} &\le A\|T^{\phi_1}_{\Theta_1} f\|_{L^1}\,\|T^{\phi_2}_{\Theta_2} g\|_{L^\infty}\\
&\lesssim A\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}\|f\|_{L^1}\|g\|_{L^\infty},
\end{aligned}$$

and by (2.10), (2.12), and (2.15)

$$\begin{aligned}\|T^{\phi_1,\phi_2}_{\tau_1}(f,g)\|_{L^1} &\le A\sum_{j\in\mathbb{N}} 2^{-jn/2}\,\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}f\|_{L^1}\,\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\varphi(\eta)}g\|_{L^\infty}\\ &\lesssim A\sum_{j\in\mathbb{N}} 2^{-jn/2}\,2^{j(\frac{n-1}{2}+\epsilon)}\|\theta_1\|_{C^N}\|f\|_{L^1}\|\theta_2\|_{C^N}\|g\|_{L^\infty}\\ &\lesssim A\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}\|f\|_{L^1}\|g\|_{L^\infty}.\end{aligned}$$

These imply the desired estimates for $\widetilde{\sigma}_0$ and τ_1 with the embedding $h^1 \hookrightarrow L^1$.

Next, we consider the amplitude τ_2. To obtain the desired inequality, it suffices to prove

$$\big\|T^{\phi_1,\phi_2}_{\tau_2}(f,g)\big\|_{L^1} \lesssim A\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}\|g\|_{L^\infty}$$

for h^1-atoms f, which satisfy that

$$\operatorname{supp} f \subset \{y\in\mathbb{R}^n \mid |y-\bar{y}|\le r\},\quad \|f\|_{L^\infty}\le r^{-n},$$

and, in addition, if $r\le 1$

$$\int f(y)\,dy = 0.$$

For the case $r\ge 1$, since $\|f\|_{L^2}\lesssim r^{-n/2}\le 1$, we have by (2.11), (2.13), and (2.17)

$$\begin{aligned}&\|T^{\phi_1,\phi_2}_{\tau_2}(f,g)\|_{L^1} \le |\widetilde{K}|^{1/2}\|T^{\phi_1,\phi_2}_{\tau_2}(f,g)\|_{L^2}\\ &\lesssim A\sum_{j\in\mathbb{N}} 2^{-jn/2}\,\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}f\|_{L^2}\,\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)}g\|_{L^\infty}\\ &\lesssim A\sum_{j\in\mathbb{N}} 2^{-jn/2}\,2^{j(\frac{n-1}{2}+\epsilon)}\|\theta_1\|_{C^N}\|f\|_{L^2}\|\theta_2\|_{C^N}\|g\|_{L^\infty}\\ &\lesssim A\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}\|g\|_{L^\infty}.\end{aligned}$$

For the case $r\le 1$, we will use the following lemma. This lemma is essentially proved in [KMT, Lemma 4.4], but we will give its proof in Appendix A for the reader's convenience.

Lemma 2.2. *Let $n\ge 2$ and $\mathcal{K}\subset\mathbb{R}^n$ be a compact set, and let ζ be the smooth function given in Notation 1.5 and let $\theta\in C^\infty_0(\mathbb{R}^n)$ satisfy $\operatorname{supp}\theta\subset\{|\xi|\le 2\}$. Suppose that ϕ is a non-degenerate phase function and set $R = 1+\sup\big\{|\nabla_\xi\phi(x,\xi)| \mid x\in\mathcal{K}, |\xi|=1\big\}$. Then for each positive integer $N>n$, there exists a positive constant C such that*

$$\left|\int_{\mathbb{R}^n} e^{-iy\cdot\xi+i\phi(x,\xi)}\zeta(\xi)\theta(2^{-j}\xi)\,d\xi\right| \le C\,\|\theta\|_{C^N}|y|^{-N}$$

for $x\in\mathcal{K}$, $|y|\ge 2R$, and $j\in\mathbb{N}$.

Now, we shall prove the case $r\le 1$. We set

$$R = 1+\sup\{|\nabla_\xi\phi_2(x,\xi)| \mid x\in\widetilde{K},\ |\xi|=1\}$$

and decompose g as $g = g\mathbf{1}_B + g\mathbf{1}_{B^c}$, where $B=\{x\in\mathbb{R}^n \mid |x|\le 2R\}$. We first consider the estimate for $T^{\phi_1,\phi_2}_{\tau_2}(f, g\mathbf{1}_{B^c})$. By the kernel representation of (linear) Fourier integral operators, we have

$$\begin{aligned}&T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)}(g\mathbf{1}_{B^c})(x)\\ &= \int_{|y|\ge 2R} g(y)\Big\{\chi(x)\int_{\mathbb{R}^n} e^{-iy\cdot\eta+i\phi_2(x,\eta)}\zeta(\eta)\theta_2(2^{-j}\eta)\,d\eta\Big\}dy.\end{aligned}$$

Then, Lemma 2.2 yields that

$$\begin{aligned}\sup_{x\in\widetilde{K}}\big|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)}(g\mathbf{1}_{B^c})(x)\big| &\lesssim \int_{|y|\geq 2R}|g(y)|\|\theta_2\|_{C^N}|y|^{-N}\,dy\\ &\lesssim \|\theta_2\|_{C^N}\|g\|_{L^\infty}.\end{aligned}$$

Therefore, since $\|f\|_{L^1}\lesssim 1$, we have by (2.11), (2.12), and the above estimate

$$\begin{aligned}\|T^{\phi_1,\phi_2}_{\tau_2}(f,g\mathbf{1}_{B^c})\|_{L^1} &\leq A\sum_{j\in\mathbb{N}}2^{-jn/2}\,\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}f\|_{L^1}\,\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)}(g\mathbf{1}_{B^c})\|_{L^\infty(\widetilde{K})}\\ &\lesssim A\sum_{j\in\mathbb{N}}2^{-jn/2}\,2^{j(\frac{n-1}{2}+\epsilon)}\|\theta_1\|_{C^N}\|f\|_{L^1}\|\theta_2\|_{C^N}\|g\|_{L^\infty}\\ &\lesssim A\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}\|g\|_{L^\infty}.\end{aligned}$$

We finally consider the estimate for $T^{\phi_1,\phi_2}_{\tau_2}(f,g\mathbf{1}_B)$. We notice from (2.13) that

$$\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}f\|_{L^2}\lesssim\|\theta_1\|_{C^N}\|f\|_{L^2}\lesssim\|\theta_1\|_{C^N}2^{jn/2}\big(2^jr\big)^{-n/2}. \tag{2.18}$$

Moreover, we have by the kernel representation

$$T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}f(x)=\int_{\mathbb{R}^n}f(y)K_j(x,y)\,dy,$$

where

$$K_j(x,y)=\int_{\mathbb{R}^n}e^{-iy\cdot\xi+i\phi_1(x,\xi)}\chi(x)\theta_1(2^{-j}\xi)\,d\xi.$$

Here, by the moment condition of h^1-atoms, we have

$$\begin{aligned}T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}f(x)&=\int_{\mathbb{R}^n}f(y)\Big\{K_j(x,y)-K_j(x,\bar{y})\Big\}\,dy\\ &=\sum_{|\alpha|=1}\iint_{\substack{|y-\bar{y}|\leq r,\\ 0<t<1}}f(y)\,(y-\bar{y})^\alpha\,\big(\partial_y^\alpha K_j\big)\big(x,\bar{y}+t(y-\bar{y})\big)\,dtdy,\end{aligned}$$

where $\bar{y}\in\mathbb{R}^n$ is the center of the ball containing $\operatorname{supp}f$. By taking a function $\widetilde{\theta}\in C_0^\infty(\mathbb{R}^n)$ such that $\widetilde{\theta}=1$ on $\operatorname{supp}\theta_1$, the kernel $\partial_y^\alpha K_j$ can be regarded as

$$\begin{aligned}\big(\partial_y^\alpha K_j\big)(x,y)&=(-i)^{|\alpha|}\int_{\mathbb{R}^n}e^{i\phi_1(x,\xi)}\chi(x)\theta_1(2^{-j}\xi)\Big\{e^{-iy\cdot\xi}\xi^\alpha\widetilde{\theta}(2^{-j}\xi)\Big\}\,d\xi\\ &=(-i)^{|\alpha|}\,T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}\big([e^{-iy\cdot\xi}\xi^\alpha\widetilde{\theta}(2^{-j}\xi)]^\vee\big)(x),\end{aligned}$$

which implies from (2.13) and Plancherel's theorem that

$$\begin{aligned}\|\big(\partial_y^\alpha K_j\big)(x,y)\|_{L^2_x}&=\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)}\big([e^{-iy\cdot\xi}\xi^\alpha\widetilde{\theta}(2^{-j}\xi)]^\vee\big)(x)\|_{L^2_x}\\ &\lesssim\|\theta_1\|_{C^N}\|[e^{-iy\cdot\xi}\xi^\alpha\widetilde{\theta}(2^{-j}\xi)]^\vee\|_{L^2}\\ &\approx 2^{jn/2}\,2^{j|\alpha|}\|\theta_1\|_{C^N}\end{aligned}$$

for any $y \in \mathbb{R}^n$. Hence, we have

$$\begin{aligned} &\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)} f\|_{L^2} \\ &\le \sum_{|\alpha|=1} \iint_{\substack{|y-\bar{y}|\le r,\\ 0<t<1}} |f(y)|\,|y-\bar{y}| \left\| \big(\partial_y^\alpha K_j\big)\big(x, \bar{y}+t(y-\bar{y})\big)\right\|_{L^2_x} dtdy \\ &\lesssim 2^{jn/2}\,(2^j r)\|\theta_1\|_{C^N}. \end{aligned} \tag{2.19}$$

Combining (2.18) and (2.19), we obtain

$$\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)} f\|_{L^2} \lesssim 2^{jn/2} \min\{(2^j r)^{-n/2}, (2^j r)\}\|\theta_1\|_{C^N}. \tag{2.20}$$

Therefore, since $\|g\mathbf{1}_B\|_{L^2} \lesssim \|g\|_{L^\infty}$, we have by (2.11), (2.20), and (2.16)

$$\begin{aligned} &\|T^{\phi_1,\phi_2}_{\tau_2}(f, g\mathbf{1}_B)\|_{L^1} \\ &\le A\sum_{j\in\mathbb{N}} 2^{-jn/2}\,\|T^{\phi_1}_{\Theta_1(x,2^{-j}\xi)} f\|_{L^2}\,\|T^{\phi_2}_{\Theta_2(x,2^{-j}\eta)\zeta(\eta)}(g\mathbf{1}_B)\|_{L^2} \\ &\lesssim A\sum_{j\in\mathbb{N}} 2^{-jn/2}\,2^{jn/2}\min\{(2^j r)^{-n/2}, (2^j r)\}\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}\|g\mathbf{1}_B\|_{L^2} \\ &\lesssim A\|\theta_1\|_{C^N}\|\theta_2\|_{C^N}\|g\|_{L^\infty}. \end{aligned}$$

This completes the proof of Proposition 2.1. □

Appendix A.

In this appendix, we prove Lemma 2.2. To this end, we will use the lemma below, which is essentially proved in [KMT, Lemma 4.3].

Lemma A.1. *Let $n \ge 2$ and $\mathcal{K} \subset \mathbb{R}^n$ be a compact set, and let ψ be a C^∞ function on $\mathbb{R}^n$ satisfying* $\operatorname{supp}\psi \subset \{2^{-1} \le |\xi| \le 2\}$. *Suppose that ϕ is a non-degenerate phase function and set $R = 1 + \sup\big\{|\nabla_\xi \phi(x,\xi)| \mid x \in \mathcal{K}, |\xi| = 1\big\}$. Then for each positive integer N, there exists a positive constant C such that*

$$\left|\int_{\mathbb{R}^n} e^{-iy\cdot\xi + i\phi(x,\xi)}\psi(2^{-j}\xi)\,d\xi\right| \le C\,\|\psi\|_{C^N}(2^j)^{n-N}|y|^{-N}$$

for $x \in \mathcal{K}$, $|y| \ge 2R$, and $j \in \mathbb{N}$.

Proof. In this proof, $I_j(x,y)$ denotes the integral of the left hand side. By changing variables,

$$I_j(x,y) = 2^{jn}\int_{\mathbb{R}^n} e^{-i2^j(y\cdot\xi - \phi(x,\xi))}\psi(\xi)\,d\xi.$$

For $j \in \mathbb{N}$ and $\ell = 1, \dots, n$, we write

$$F_j = 2^j\big(y\cdot\xi - \phi(x,\xi)\big), \quad G_{j,\ell} = \frac{\partial_{\xi_\ell} F_j}{|\nabla_\xi F_j|^2}.$$

Then, since

$$e^{-iF_j} = i\sum_{\ell=1}^n \frac{\partial_{\xi_\ell} F_j}{|\nabla_\xi F_j|^2}\,\partial_{\xi_\ell} e^{-iF_j} = i\sum_{\ell=1}^n G_{j,\ell}\,\partial_{\xi_\ell} e^{-iF_j},$$

we have by integration by parts

$$I_j(x,y) = 2^{jn}\int e^{-iF_j}\psi\,d\xi = 2^{jn}\int \Big(i\sum_{\ell=1}^{n} G_{j,\ell}\,\partial_{\xi_\ell} e^{-iF_j}\Big)\psi\,d\xi$$
$$= 2^{jn}\sum_{\ell=1}^{n}(-i)\int e^{-iF_j}\,\partial_{\xi_\ell}\big(G_{j,\ell}\,\psi\big)\,d\xi.$$

Repeating integration by parts N-times, we have

$$I_j(x,y) = 2^{jn}\sum_{\substack{\ell_1,\dots,\ell_N\in\{1,\dots,n\},\\ |\alpha_1+\cdots+\alpha_N+\beta|=N}} c\int e^{-iF_j}\big(\partial_\xi^{\alpha_1}G_{j,\ell_1}\big)\dots\big(\partial_\xi^{\alpha_N}G_{j,\ell_N}\big)\big(\partial_\xi^{\beta}\psi\big)\,d\xi.$$

Now, we observe that, for $|\alpha|\geq 2$,

$$|\nabla_\xi F_j| = 2^j|y-\nabla_\xi\phi(x,\xi)| \approx 2^j|y|,$$
$$|\partial_\xi^\alpha F_j| = 2^j|\partial_\xi^\alpha\phi(x,\xi)| \lesssim 2^j \leq 2^j|y|,$$

for $x\in\mathcal{K}$, $|y|\geq 2R$, and $1/2\leq|\xi|\leq 2$. Then, we see that for any $\alpha\in(\mathbb{N}_0)^n$

$$|\partial_\xi^\alpha G_{j,\ell}| \lesssim (2^j|y|)^{-1}$$

holds. Therefore, we obtain

$$|I_j(x,y)| \lesssim 2^{jn}\int_{1/2\leq|\xi|\leq 2}(2^j|y|)^{-N}\|\psi\|_{C^N}\,d\xi$$
$$\approx \|\psi\|_{C^N}(2^j)^{n-N}|y|^{-N},$$

which completes the proof. □

Proof of Lemma 2.2. From the property of ζ stated in Notation 1.5, we have

$$\zeta(\xi)\theta(2^{-j}\xi) = \sum_{k=1}^{j+1}\psi(2^{-k}\xi)\theta(2^{-j}\xi).$$

If $|y|\geq 2R$ and $1\leq k\leq j+1$, then Lemma A.1 gives

$$\left|\int_{\mathbb{R}^n} e^{-iy\cdot\xi+i\phi(x,\xi)}\psi(2^{-k}\xi)\theta(2^{-j}\xi)\,d\xi\right| \lesssim (2^k)^{n-N}|y|^{-N}\left\|\psi(\cdot)\theta(2^{k-j}\cdot)\right\|_{C^N}$$
$$\lesssim (2^k)^{n-N}|y|^{-N}\left\|\theta\right\|_{C^N}.$$

Hence, if $N>n$, then taking sum over k, we complete the proof. □

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